EE4-16
EE9-AC6

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

MSc and EEE PART IV: MEng and ACGI

Corrected Copy

ANALOGUE SIGNAL PROCESSING

Friday, 17 May 10:00 am

Time allowed: 3:00 hours

There are SIX questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : E. Rodriguez-Villegas

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Special instructions for students

Unless otherwise stated the following parameters have the following definitions:

V_{DS}*: Drain Source Voltage.*

V_{GS}*: Gate Source Voltage.*

V_{TH}*: Threshold Voltage.*

U_t*: Thermal Voltage.*

n *: Weak Inversion Slope factor.*

g_m*: Transconductance.*

1.

a) State two advantages and two disadvantages of encoding analogue signals in time rather than amplitude.

[4]

b) The circuit shown in Figure 1.1 is based on two phases, a reset phase when RST = 0 and a readout phase where RST = V_{DD}. The input voltage is V_{in}.

i. State which domain the output V_{out} is encoded in and what limits the dynamic range of the encoded output signal.

[2]

ii. Assuming that M0 is biased in weak inversion, describe the behavior of V_a in both phases. Derive the rate of change of V_a during the readout phase. Considering that the switching point of the inverters is at $\frac{V_{DD}}{2}$, derive the transfer function between the encoded output and V_{in}. Define all constants used in MOSFET transconductance equations.

[6]

c) For the cascade of amplifiers shown in Figure 1.2, derive the output referred noise and explain what factors can be optimized to improve the signal to noise ratio.

[8]

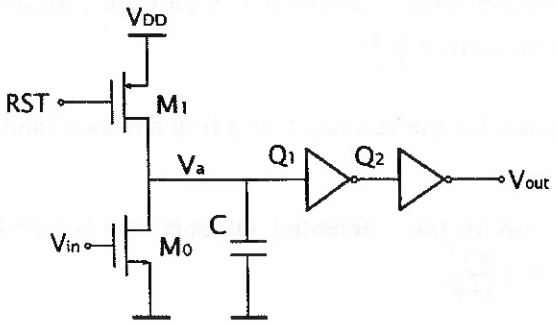


Figure 1.1

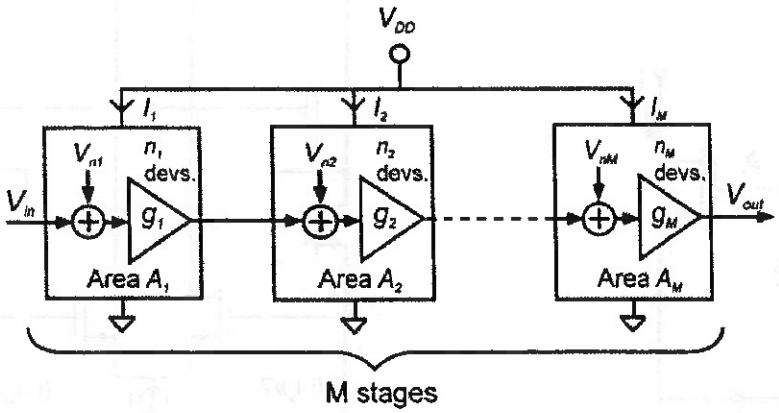


Figure 1.2

2. This question concerns translinear circuits.

a) Give two limitations when using MOSFETs in translinear circuits.

[2]

b) Figure 2.1 shows a translinear circuit.

i. Derive the function of the circuit using the translinear principle. I_z is the output and I_x is the input.

[2]

ii. Show how, with the addition of a few transistors, the circuit can be modified to calculate the function:

$$I_Z = \frac{(I_X)^4}{(I_0)^3} - I_X$$

[4]

iii. A capacitor is added between the point A on the circuit in Figure 2.1 and ground. Derive the new transfer function of the circuit stating the gain and time constant. The final equation of the transfer function should be represented in the s-domain.

[4]

c) Figure 2.2 shows a translinear circuit capable of computing the trigonometric function $\sin(\pi y)$ through the approximation $\frac{y-y^3}{1+y^2}$.

i. Derive an expression for the currents I_3 and I_4 in terms of I and I_X.

[4]

ii. Derive an expression for the differential output current I_2-I_1 and show that when $I_X = yI$ then $I_2 - I_1 = I\frac{y-y^3}{1+y^2}$

[4]

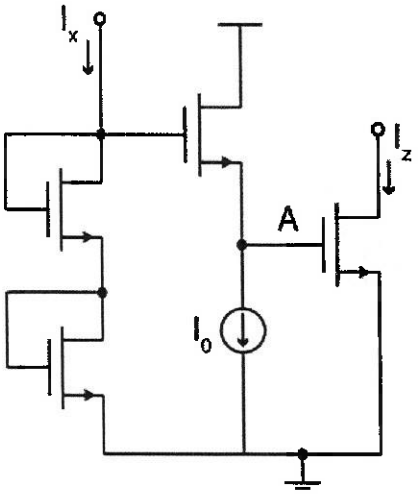


Figure 2.1

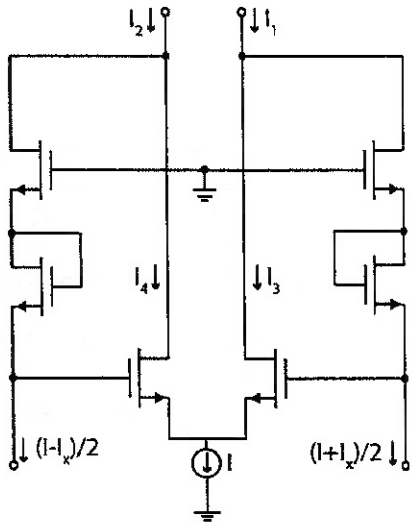


Figure 2.2

3.

a) Explain the principle of companding, illustrating how the dynamic range changes through the signal processing chain.

[4]

b) Figure 3.1 shows a block diagram of a current-mode companding integrator, with an input X and an output Y. f() and g() represent non-linear functions which compress and expand the signal such that the input-output relationship of the integrator is linear and given by:

$$Y = \tau \int X \cdot dt$$

Given that the expansive function is defined as $I_{out} = g(V_c) = I_0 \sinh(V_c/nU_t)$, derive the function f(x) such that linear integration is achieved. All constants should be grouped to represent a current I_1, with the function f(x) composed of just I_{in}, I_1 and I_{cosh}, whereby $I_{cosh} = I_0 \cosh(V_c/nU_t)$.

[6]

c) The transfer function of a biquad filter is defined in state space representation by the following equations:

$$\dot{X}_1 = \frac{-\omega_0}{Q} X_1 - \omega_0 X_2 + \omega_0 U_1$$

$$\dot{X}_2 = \omega_0 X_1 - \omega_0 U_2$$

$$Y = X_1$$

whereby Y is the output and U_1 and U_2 are the inputs and X_1 and X_2 are the states.

i. By using the mappings below show how these linear equations can be mapped to non-linear log-domain design equations which represent a low pass filter. State any assumptions you make.

$$X_1 = I_1 \exp\left(\frac{V_1}{nU_t}\right) \qquad X_2 = I_2 \exp\left(\frac{V_2}{nU_t}\right) \qquad U_2 = I_{U2} \exp\left(\frac{V_{U2}}{nU_t}\right)$$

[5]

ii. With these log-domain design equations, sketch a schematic of the final log domain filter using weak inversion MOS transistors.

[5]

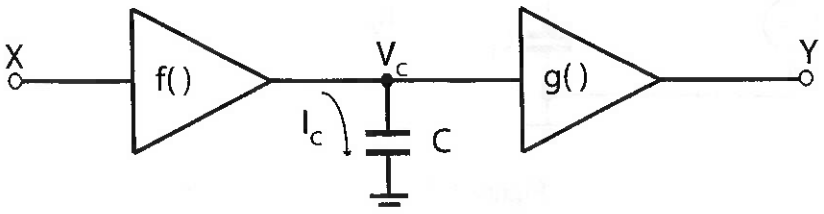


Figure 3.1

4.

a) Figure 4.1 shows a conventional second-generation positive current conveyor, CCII+.

i. Explain its operation principle and describe how impedances at each of its three ports, X, Y, Z, differ from a standard operational amplifier.

[4]

ii. Draw the circuit of a CMOS implementation of a bi-directional CCII+ with high output impedance.

[4]

iii. Using the current conveyor in Figure 4.1, draw the schematics of a current mode integrator and a voltage mode integrator, each with a time constant, τ =1s. You may assume R = 100KΩ.

[4]

b) Figure 4.2 shows a voltage mode Sallen key filter. Using the adjoint principle transform this to its current mode equivalent using CCII+ blocks.

[4]

c) Sketch a current mode circuit that can implement the following hyperbolic functions:

$$I_1 - I_2 = I_{DC} \sinh\left(\frac{V_1 - V_2}{nU_t}\right)$$

$$I_1 + I_2 = I_{DC} \cosh\left(\frac{V_1 - V_2}{nU_t}\right)$$

[4]

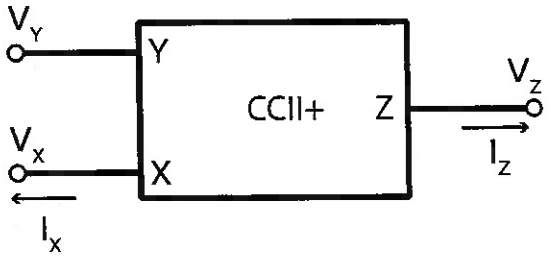


Figure 4.1

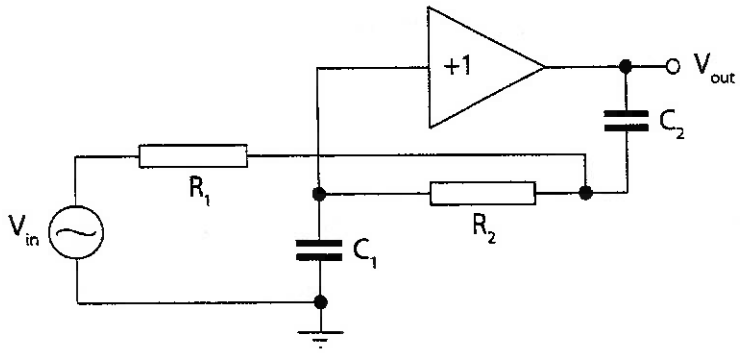


Figure 4.2

5. This question relates to discrete time integrators.

a) Figure 5.1 shows a discrete time RC integrator.

i. Derive the time constant of integration. Assuming $C_1 = C_2$, select a suitable clock frequency to give a time constant $\tau = 1\mu s$.

[2]

ii. State four possible non-idealities in switches. Describe a way to reduce the one which is usually dominant.

[4]

iii. For the circuit Figure 5.1, when CK goes low all of the channel charge, Q_1, of transistor M_1 is injected into C_1. This causes an error voltage ΔV_{C1} in C_1 defined by:

$$\Delta V_{C1} = \frac{Q_1}{C_1}$$

Then transistor M_2 turns ON. The channel charge of M_2 is split equally between C_1 and C_2. Estimate the total amount of charge injection and therefore error voltage ΔV_{C2} from switches M_1 and M_2 which appears on the output capacitor C_2 after CK goes high again, that is after one integration cycle.

[4]

b) Figure 5.2 shows an autozeroing switched capacitor integrator capable of removing any offset in the amplifier. Derive the output voltage of the integrator during phase Φ_1 and Φ_2 indicating how the offset is removed by showing the charge on each capacitor during each phase.

[5]

c) Figure 5.3 shows a schematic of switched current integrator.

i. Derive the transfer function of the switched current integrator and show that in the z-domain this may be represented as H(z) = A/(z-1).

[5]

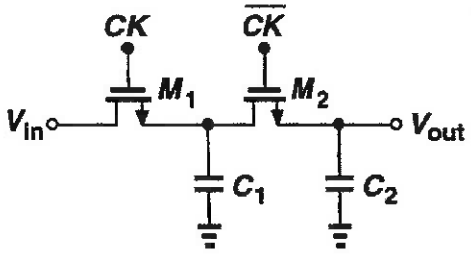


Figure 5.1

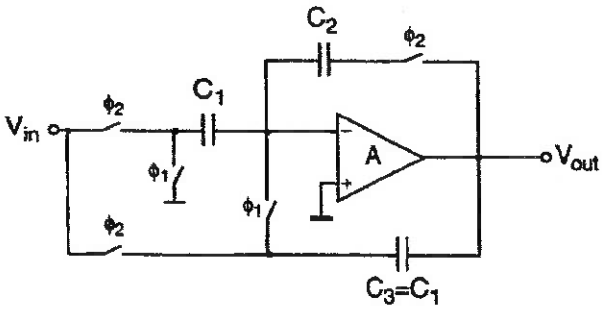


Figure 5.2

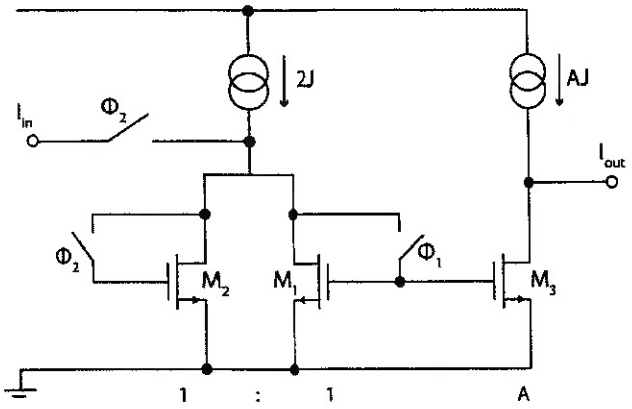


Figure 5.3

6. Figure 6.1 shows a measurement system that digitizes the product of two different signals. The architecture uses an operational amplifier, a current multiplying circuit and an analogue-to-digital converter to encode the detected voltage signal. The detected signals are first converted to currents through a voltage-to-current converter (V/I), then multiplied in current mode and finally converted to a spike train through an integrate-and-fire neuron (I/F). The 8 bit counter is used to count the number of spikes in a fixed period T_{reset}. The whole system works off a 1 V supply.

a) Describe a technique which allows detection of very small voltage signals which exist below the 1/f noise floor of the operational amplifier. Include diagrams showing the operation of this technique and how the frequency spectrum of the signal and noise is shaped in each stage.

[5]

b) Sketch a circuit for multiplying two currents. State the region of operation for the MOSFETs and derive the relationship between the output and the input. Identify a way to provide a gain to the multiplication.

[4]

c) Sketch an integrate-and-fire circuit which can be used in this design and describe its operation. Derive equations for the pulse width and frequency.

[5]

d) Given that the maximum output is limited to $I_{out} = 100$ nA, choose a suitable capacitor value for your integrate-and-fire circuit such that the spike frequency is always less than 5 kHz. You may assume the refractory current is $I_r = 1$ nA.

[4]

e) Calculate the maximum allowable reset time T_{rst} such that the counter never overflows for this current. How many bits are required when integrating over a long time constant $T_{rst} = 1\ s$?

[2]

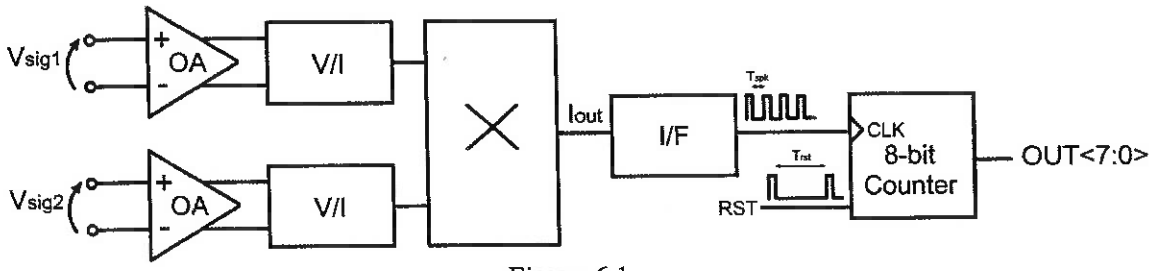


Figure 6.1